

Supplementary Tables

Supplementary Table 1. Parameter settings of evaluated LLMs for GPBench.

Model	Temperature	Top-p	Repetition Penalty
GPT-4-turbo	1	1	0
o1-preview	1	1	0
Gemini-1.5-pro	Not Reported	0.95	Not Reported
Qwen2.5-7B-Instruct	0.7	0.7	0.5
Qwen2.5-72B-Instruct	0.7	0.7	0.5
Claude-3.5-sonnet	1	Not Reported	Not Reported
DeepSeek-V3	1	Not Reported	Not Reported
DeepSeek-R1	1	Not Reported	Not Reported
HuatuoGPT-o1-7B	0.3	0.8	1.05

Supplementary Table 2. Specializations and Experience Levels of Experts.

Members	Affiliation	Specialization
Jiping Lang (MD)	The Sixth Affiliated Hospital of Sun Yat-sen University	General practice
Shuang Li (M.Med)	The Sixth Affiliated Hospital of Sun Yat-sen University	General practice
Xuanna Li (M.Med)	The Sixth Affiliated Hospital of Sun Yat-sen University	General practice
Yuze Tang (M.Med)	The Sixth Affiliated Hospital of Sun Yat-sen University	General practice
Zhu Lin (M.Med)	The Sixth Affiliated Hospital of Sun Yat-sen University	General practice
Xiaolin Lu (MBBS)	Xinyi People's Hospital	Cardiology
Hongji Yu (M.Med)	Xinyi People's Hospital	Endocrinology
Jiexian Qiu (MBBS)	Xinyi People's Hospital	Cardiovascular medicine
Junrong Chen (M.Med)	The Sixth Affiliated Hospital of Sun Yat-sen University	General practice
Lin Yao (M.Med)	The Sixth Affiliated Hospital of Sun Yat-sen University, Xinyi People's Hospital	General practice
Wenhao Jiang (Ph.D.)	Guangdong Laboratory of Artificial Intelligence and Digital Economy (SZ)	Artificial Intelligence
Zheqing Li (MSc)	The Sixth Affiliated Hospital of Sun Yat-sen University	Electronic Engineering

Supplementary Table 3. An example of annotation from the *MCQ Test Set*.

Field	Value
ID	1278
Question	Female, 40 years old, with a 10-year history of rheumatoid arthritis, has been taking generic nonsteroidal anti-inflammatory drugs (NSAIDs) for a long period. Laboratory tests show a hemoglobin level of 78 g/L in the complete blood count. Which of the following statements regarding this patient's anemia is incorrect?
Options	A: It is microcytic hypochromic anemia. B: It belongs to anemia of chronic disease. C: The primary mechanism is impaired iron utilization. D: Iron deficiency factors may be involved. E: It is often accompanied by thrombocytopenia.
Answer	E
Type	single-answer multiple-choice question
Source	cmb-test - Physician Exam - Intermediate-Level Title - Attending Physician of Internal Medicine
Indicators	Diagnosis and Differential Diagnosis (I2-1), Complication Identification (I2-3), Adverse Drug Reaction Management (I3-2)

Supplementary Table 4. The scoring criteria for the competency in medical history taking on the *AI Patient Test Set*

Primary Dimension			Secondary Dimension	Score
General points)	Information	(5	Full and accurate documentation of gender, age, occupation, and other non-identifiable demographic information.	5
History of Present Illness (50 points)			Onset of disease: time of onset, acuity of progression, potential etiological factors or predisposing causes.	6
			Characteristics of the main symptoms or signs: location, nature, duration, severity, alleviating and exacerbating factors.	10
			Disease progression and clinical course.	6
			Accompanying symptoms.	6
			Symptoms relevant for differential diagnosis.	6
			Diagnosis and treatment process: whether the patient has sought medical attention; examinations performed; medications or treatments prescribed, and their effectiveness.	10
Relevant Medical History (15 points)			General condition and related current medical history.	6
			Past medical history.	3
			Personal history.	3
			Menstrual history (for pediatrics: birth history).	3
			Marital and reproductive history (for pediatrics: growth and developmental history).	3
Clinical Interviewing Skills (20 points)			Family history.	3
			Clear structure, with a focus on key points.	7
			Focus on the patient's condition, using simple and understandable language.	7
Professionalism (10 points)			Non-suggestive, non-leading, and non-accusatory questioning.	6
			Good service attitude, caring and considerate towards patients.	5
			Strong communication skills, skillful guidance, no abrupt interruptions of patient narration.	5

Supplementary Table 5. An example of *Fever* outpatient medical record.

Case
Patient: male, 47 years old. Chief Complaint: Fever for one week. Present Illness: One week ago, the patient began to have a fever of T 38°C after mountain climbing, accompanied by chills and rigors, as well as coughing and expectoration of copious white viscous sputum. He also experienced muscle aches. A complete blood count performed at a local clinic indicated decreased platelets. After treatment, there was no significant improvement in symptoms; the fever recurred with a body temperature rising to T 39°C along with rigors, dizziness, headache, and occasional nausea. He came to our hospital for further evaluation. Past Medical History: He has had stage 2 hypertension for many years. He is currently not taking any antihypertensive medications, and his blood pressure has remained within normal limits. Physical Examination: T 38°C, P 120 beats/min, R 28 breaths/min, BP 136/76 mmHg. Conscious and alert. A 0.5 cm × 0.5 cm skin ulceration was observed on the left lower limb. Bilateral lung sounds were slightly diminished, with a small amount of moist rales heard in both lower lungs. The heart rate was 120 beats/min, regular rhythm, and no obvious pathological murmurs were detected in any valve area. The abdomen was flat and soft, with no tenderness or rebound tenderness. The liver and spleen were not palpable below the costal margin. Muscle strength and tone in all four limbs were normal, and no edema was noted in either lower extremity. Laboratory and Auxiliary Examinations: Complete blood count: WBC: $8.96 \times 10^9/L$, NEUT: 50.6%, Hb: 150.0 g/L, PLT: $48 \times 10^9/L$(↓). D-dimer: 13.06 μg/mL(↑), BNP: 69.90 pg/mL(↑), hs-cTnI: 0.033 ng/mL, tests for respiratory pathogens, hepatitis B serological markers, and transfusion-transmitted infections were normal. Biochemistry: normal cardiac enzymes, BUN: 6.33 mmol/L, Cr: 81.50 μmol/L, Ca: 1.95 mmol/L(↓), hs-CRP: 73.52 mg/L(↑), ALT: 190.64 U/L(↑), AST: 203.98 U/L(↑), γ-GT: 173.50 U/L(↑), ALB: 28.50 g/L(↓), FER: > 1025.0 ng/mL(↑). ECG : (1) Sinus rhythm. (2) Low voltage in limb leads. Cardiac ultrasound: mild mitral regurgitation, mild tricuspid regurgitation, normal left ventricular systolic function. Abdominal and urinary tract ultrasound: slight splenomegaly; no obvious abnormalities in the liver, gallbladder, bile ducts, pancreas; mildly enlarged prostate; kidneys within normal size range; no ureteral dilation; no abnormal findings in the bladder. Chest CT report: (1) Multiple scattered inflammatory and fibrotic lesions in both lungs, with a small amount of bilateral pleural effusion. Further follow-up is recommended. (2) Bilateral pleural thickening; possible encapsulated pleural effusion in the right interlobar region, to be correlated clinically. Diagnosis: 1. Scrub typhus 2. Splenomegaly 3. Liver dysfunction 4. Pleural effusion 5. Hypoproteinemia 6. Secondary thrombocytopenia 7. Stage 2 hypertension, high-risk category Treatment: I. General Management The patient should rest, eat easily digestible foods, maintain oral hygiene, and ensure water and electrolyte balance to reduce complications. For high fever, physical cooling or appropriate use of antipyretic agents may be employed, avoiding excessive sweating caused by certain antipyretics. II. Etiological Treatment (select one of the following) 1. Tetracyclines: Recommended dose of doxycycline for adults is 0.1 g twice a day; the first dose is doubled. Continue the anti-infective therapy for at least 3 days after body temperature returns to normal and clinical symptoms improve and stabilize, generally for 7—10 days. In severe cases, intravenous administration of doxycycline may be considered: on day 1, 0.2 g in 1—2 infusions, then 0.1—0.2 g daily depending on infection severity. 2. Macrolides: Roxithromycin 150 mg twice a day for adults; after fever subsides, 150 mg once a day. Azithromycin 0.5 g once daily for adults; after fever subsides, 0.25 g once daily. Clarithromycin 0.5 g once every 12 hours for adults. All of these regimens are given for 7—10 days. III. Symptomatic and Supportive Treatment In cases of scrub typhus complicated by respiratory, hepatic, or renal dysfunction, treatments may include anti-inflammatory therapy, oxygen supplementation, liver protection and enzyme-lowering measures, as well as maintenance or replacement therapy for renal function.

Supplementary Table 6. An example of annotated scoring criteria for the sample of *Fever* medical record.

Indicator	Applicability		Scoring Criteria
Diagnosis and Differential Diagnosis	Yes		1. Scrub typhus (40 points) 2. Splenomegaly (10 points) 3. Hepatic insufficiency (10 points) 4. Pleural effusion (10 points) 5. Hypoproteinemia (10 points) 6. Secondary thrombocytopenia (10 points) 7. Grade 2 hypertension; high-risk group (10 points)
Referral Decision-making	Yes		Referral to the department of infectious diseases (100 points).
Acute and Critical Condition Recognition	Yes		Hepatic insufficiency (25 points); Pleural effusion (25 points); Hypoproteinemia (25 points); Secondary thrombocytopenia (25 points)
Complication Identification	Yes		No critical or severe conditions were involved (100 points).
Formulation of Optimal Treatment Plan	Yes		Part 1. Optimal treatment plan I. Etiological Treatment (choose one medication if appropriate) (50 points; if the medication name or usage and dosage are incorrect, deduct 30 points) 1. Tetracyclines: Doxycycline Recommended dosage for adults: 0.1 g twice a day, with the initial dose doubled. The anti-infective course should continue for at least 3 days after body temperature returns to normal and clinical symptoms have improved and stabilized, generally for 7–10 days. For severe cases, intravenous infusion may be selected. On the first day, adults receive 0.2 g in 1–2 infusions. Thereafter, according to the severity, 0.1–0.2 g is administered intravenously per day. 2. Macrolides: Roxithromycin 150 mg twice a day for adults; once the fever subsides, 150 mg once a day. Azithromycin 0.5 g once a day for adults; after defervescence, 0.25 g once a day. Clarithromycin 0.5 g per dose for adults, administered orally once every 12 hours. All of these medications are given for a total of 7–10 days. II. Symptomatic and Supportive Treatment (50 points) When scrub typhus is complicated by respiratory, hepatic, or renal dysfunction, measures often include anti-inflammatory therapy, oxygen supplementation, liver-protective interventions with enzyme-lowering therapy, and maintenance or replacement of renal function. Part 2. Clinical appropriateness evaluation of diagnostic examinations N/A
Health Education Delivery	Yes		Patients should ensure adequate rest, consume easily digestible foods, pay attention to oral hygiene, maintain fluid and electrolyte balance, and reduce the risk of complications. Physical cooling measures may be used for high fever, and antipyretic medications can be administered as appropriate. However, caution should be exercised in the use of fever-reducing drugs that induce excessive sweating(100 points).
Cost-effective Coordination	Care	No	N/A
Humanistic Competency	Care	No	N/A

Supplementary Table 7. The accuracies (%) of LLMs on the *MCQ Test set* across the secondary competency indicators.

Primary Indicator	Secondary Indicator	Qwen2.5-7B-Instruct	Qwen2.5-72B-Instruct	GPT-4o	GPT-4-turbo	o1-preview	Claude-3.5-Sonnet	Gemini-1.5-pro	HuatuogPT-o1-7B	DeepSeek-V3	DeepSeek-R1
I1. Medical Knowledge (10%)	I1-1. Basic Medical Knowledge (100%)	64.48	76.60	68.83	54.53	80.01	53.73	64.98	70.59	71.86	82.44
	I2-1. Diagnosis and Differential Diagnosis (20%)	68.90	79.80	74.36	60.86	82.18	59.36	68.80	74.28	77.00	85.38
I2. Diagnosis (40%)	I2-2. Medical History Taking (20%)	71.82	79.82	77.45	59.27	85.45	61.09	68.55	74.18	80.00	85.82
	I2-3. Complication Identification (20%)	65.95	80.83	73.39	62.66	83.12	55.65	69.38	73.24	78.97	89.27
	I2-4. Acute and Critical Condition Recognition (20%)	77.68	82.90	82.32	63.77	86.38	59.13	69.28	82.03	82.03	87.54
	I2-5. Referral Decision-making (20%)	65.82	70.89	64.56	72.15	62.03	72.78	73.42	60.51	68.35	61.39
I3. Decision-making (30%)	I3-1. Formulation of Optimal Treatment Plan (35%)	68.35	76.88	65.35	51.75	74.62	50.65	61.10	72.12	73.88	82.58
	I3-2. Adverse Drug Reaction Management (25%)	66.86	77.71	69.71	58.86	75.43	48.57	63.43	75.43	74.86	80.57
	I3-3. Contraindications Awareness (25%)	59.31	74.83	69.66	46.55	81.38	48.97	64.48	67.59	76.55	82.76
	I3-4. Selecting Alternative Therapy (15%)	64.12	74.12	63.53	54.71	72.35	50.00	60.00	66.67	71.18	79.12
I4. Health Management (10%)	I4-1. Health Education Delivery (50%)	74.34	78.11	77.36	72.08	86.04	68.68	76.23	76.89	76.60	84.53
	I4-2. Tertiary Prevention Implementation (50%)	79.29	81.07	72.86	67.14	86.07	57.50	73.21	83.57	76.07	89.64
I5. Health Economics (5%)	I5-1. Cost-effective Care Coordination (100%)	79.00	84.00	76.00	74.00	80.00	71.00	73.00	81.82	80.00	89.00
I6. Medical Ethics and Humanities (5%)	I6-1. Humanistic Care Competency (100%)	80.14	78.49	73.29	67.61	75.41	67.38	78.72	86.76	75.18	82.03
	Weighted Average STD	69.63 1.40	78.13 1.28	71.80 1.38	60.79 1.45	79.16 1.27	58.13 1.47	68.22 1.41	73.95 1.34	75.80 1.32	82.74 1.20

Supplementary Table 8. The performance (scores) of LLMs on the *Clinical Case Test Set*.

Model	Diagnosis and Differential Diagnosis (I2-1)	Complication Identification (I2-3)	Acute and Critical Condition Recognition (I2-4)	Referral Decision-making (I2-5)	Formulation of Optimal Treatment Plan (I3-1)	Health Education Delivery (I4-1)	Weighted Average	STD
Qwen2.5-7B-Instruct	63.36	62.00	80.86	75.71	56.60	82.07	68.63	16.26
Qwen2.5-72B-Instruct	67.57	60.36	88.14	74.14	66.43	87.71	72.79	17.18
GPT-4o	70.71	73.79	87.14	81.43	64.81	85.21	76.02	19.29
GPT-4-turbo	62.13	67.29	83.71	78.86	60.11	81.71	71.07	20.13
Claude-3.5-Sonnet	69.21	73.00	85.00	77.57	62.06	79.50	73.42	18.33
Gemini-1.5-pro	76.36	72.57	85.00	78.86	65.93	87.43	76.46	20.14
DeepSeek-V3	63.14	69.29	83.57	74.00	59.79	84.00	70.90	21.93
DeepSeek-R1	78.93	76.57	94.29	79.84	77.04	87.93	81.80	16.92
HuatuoGPT-o1-7B	52.79	70.29	87.86	72.14	60.00	89.43	70.35	19.88
o1-preview	67.64	68.86	85.71	85.29	61.09	85.00	74.24	20.26

Supplementary Table 9. The results of human physicians and LLMs on the evaluation dataset.

Human Physicians / Models	MCQ Test Set (accuracy %)	Clinical Case Test Set (score)	AI Patient Test Set (score)
Group A	62.97	87.10	77.75
Group B	62.39	85.77	76.59
Group C	68.32	81.16	61.30
GPT-4o [1]	71.80	76.02	51.09
GPT-4-turbo	60.79	71.07	42.06
o1-preview [2]	79.16	74.24	55.20
Gemini-1.5-pro [3]	68.22	76.46	47.10
Qwen2.5-7B-Instruct [4]	69.63	68.63	37.13
Qwen2.5-72B-Instruct [4]	78.13	72.79	44.01
Claude-3.5-Sonnet	58.13	73.42	53.67
DeepSeek-V3 [5]	75.80	70.90	37.14
DeepSeek-R1 [6]	82.74	81.80	-
HuatuoGPT-o1-7B [7]	73.95	70.35	36.33

Supplementary Methods

Supplementary Method 1. Prompt for evaluation on the Clinical Case Test Set

Diagnostic Plan Prompt

Role: You are a general practitioner.

Skills: Possess professional knowledge, excellent communication skills, and attention to detail. Capable of identifying referral needs, recognizing critical and severe conditions, identifying complications, and making accurate diagnoses.

Task: Based on the medical record, physical examination results and auxiliary examination results, provide a diagnosis for the patient.

Patient Medical Record: [Patient Medical Record]

Physical Examination Results: [Physical Examination Results]

Auxiliary Examination Results: [Auxiliary Examination Results]

Requirements:

- The doctor needs to determine whether the patient requires a referral and specify the department for referral; determine whether the condition is critical or severe; identify any complications; and provide a detailed and specific diagnosis.
- The output must follow the specified format. The diagnostic content should be described in words.
- Different pieces of information in the examination results should be separated by semi-colons.
- Based on the symptoms and medical history provided by the patient, diagnose the condition in detail and with specificity.
- The diagnosis should specify the type and severity level of the condition.
- If a specific diagnosis cannot be made based on the available information, provide suspected conditions.
- The severity and stage of the condition should be represented using Arabic numerals.
- Output only in JSON format, no additional content.

Treatment Plan Prompt

Role: You are a general practitioner.

Skills: Possess professional knowledge, excellent communication skills, and attention to detail. Capable of providing optimal treatment plans, recommending further examinations, offering health education, and ensuring cost-effective treatment.

Task: Based on the medical record, auxiliary examination results, physical examination results, and diagnosis, provide a decision-making plan for the patient.

Patient Medical Record: [Patient Medical Record]

Auxiliary Examination Results: [Auxiliary Examination Results]

Physical Examination Results: [Physical Examination Results]

Diagnosis: [Diagnosis]

Requirements:

- The output must follow the specified format.
- The decision-making plan should be divided into three parts: optimal treatment plan, further examinations, and health education.
- The content of the optimal treatment plan, further examinations, health education, and cost-effective treatment should be described in words.
- Different pieces of information in the optimal treatment plan, further examinations, health education, and cost-effective treatment should be separated by semicolons.
- The optimal treatment plan should include specific drug names, dosages, and the purpose of drug therapy.
- Output only in JSON format, no additional content.

Supplementary Method 2. Prompt for evaluation on the AI Patient Test Set

Patient Agent prompt

Role: You are a patient visiting a hospital.

Patient Information: [Patient Information]

Task: Respond to doctor's questions strictly according to the following categories and action guidelines to simulate realistic doctor-patient dialogue.

Goal: Make the conversation as close as possible to real patient-doctor interactions.

Question Types and Action Guidelines:

-**Initialization:** Doctor's first question. Only briefly answer about your most troubling symptoms based on the "Chief Complaint" in your personal information.

-**Conclusion:** When the doctor makes a diagnosis or reaches maximum dialogue length, remain silent and wait for further instructions.

-**Valid Inquiries:** If the doctor asks for information in your profile (including "Chief Complaint", "Present Illness History", "Past Medical History", etc.), provide the corresponding information directly and concisely.

-**Invalid Inquiries:** For any questions not covered in your profile, always answer "I'm not sure."

-**Valid Suggestions:** If the doctor recommends tests or examinations with results included in your profile, confirm and respond with those results.

-**Invalid Suggestions:** For suggested tests or examinations not in your profile, express willingness to follow these recommendations.

Special Notes:

-Keep responses brief and clear, do not repeat doctor's suggestions or conclusions.

-Use natural, conversational language to make dialogue more authentic.

-Do not mention any personal information unless specifically asked.

-Answers should directly address the doctor's questions, avoid providing irrelevant information.

-Do not mention symptoms or conditions not listed in your personal information.

-Do not mention anything not included in your personal information.

-When uncertain about an answer, consistently respond with "I'm not sure"

-Do not fabricate any false examination results.

Once the consultation begins, please respond to each of the doctor's questions one by one.

Doctor Agent prompt

Role: You are a general practitioner.

Skills: General practitioners possess the ability to diagnose and differentiate diagnoses, collect medical histories, identify complications, recognize critical and severe conditions, determine the need for referrals, provide optimal treatment plans, manage adverse drug reactions, suggest alternative treatments, educate patients on health topics, implement three levels of prevention, enhance patient compliance, offer family support, control treatment costs reasonably, and provide humanistic care.

Task: Conduct a medical consultation.

Objective: Gather as much information as possible regarding the patient's chief complaint, present illness, past medical history, and family history, and provide necessary, detailed, and accurate physical examinations, auxiliary tests, diagnoses, and treatment plans.

Consultation Steps:

Step 1: Communicate with the patient to obtain their chief complaint, present illness, past medical history, and family history.

Step 2: Once sufficient information is collected, recommend the necessary physical examination items. Physical examination refers to the process of assessing the patient's physical condition through observation, palpation, percussion, and auscultation to identify potential physical abnormalities.

Step 3: After obtaining the results of the physical examination, recommend the necessary auxiliary tests based on the patient's chief complaint, present illness, past medical history, family history, and physical examination findings. Auxiliary tests include laboratory tests (e.g., blood tests, urine tests), imaging studies (e.g., X-rays, CTs, MRIs), or other specialized diagnostic methods to gather detailed information about the patient's internal condition for diagnosis or disease evaluation.

Step 4: After obtaining the auxiliary test results, provide a detailed and accurate diagnosis and treatment plan based on the patient's chief complaint, present illness, past medical history, family history, physical examination findings, and auxiliary test results.

Requirements:

- Use conversational and simple language; keep questions short and clear.
- Do not output questions, examination items, diagnoses, and treatment plans simultaneously.
- Use layman's terms for initial diagnoses, avoiding professional jargon.
- Avoid repeating the patient's information in your questions.
- Collect details about the patient's condition step by step.
- Provide an initial diagnosis only after gathering sufficient information.
- Answer any questions the patient may have.
- Provide examination items first, followed by diagnoses and treatment plans.
- Only provide diagnoses and treatment plans after receiving examination results.
- Diagnose the patient's condition in detail and specify the type and severity of the illness.
- If a specific diagnosis cannot be made based on the available information, provide a list of suspected conditions.
- Treatment plans should include further examinations, referrals, medication therapy, and non-drug therapies.
- For medication therapy, specify the drug name, dosage, and purpose of the treatment.
- Ask only one question at a time.
- Areas not covered in the patient's physical examination and auxiliary tests are considered normal.

Special Notes:

- If the patient provides insufficient test results, do not continue asking for further test results. Instead, provide an initial diagnosis and treatment plan based on the current information and recommend additional tests to be completed.

Consultation Process:

1. Start the consultation by saying: "Hello, what seems to be the problem?"
2. When you believe you have enough information to recommend physical examination items, provide them with the phrase: "The following physical examinations are necessary:"
3. When you believe you have enough information to recommend auxiliary tests, provide them with the phrase: "The following auxiliary tests are necessary:"
4. When you believe you have enough information to make a detailed diagnosis, provide it with the phrase: "The following is the diagnosis and treatment plan:"
5. After providing the diagnosis and treatment plan, conclude with: "Consultation ended."

Supplementary Method 3. Model evaluation settings and development set design

Model inference settings and prompt design

We employed the default parameter settings for each LLM, as LLMs demonstrate strong cross-task generalization capabilities and robustness to hyperparameters, making extensive hyperparameter tuning largely unnecessary. Tuning the hyperparameters of each model for each task was prohibitively expensive due to the substantial manual annotation required. Moreover, our objective was to assess the performance of LLMs in realistic work scenarios of GPs rather than to compare the models themselves. To ensure the validity of the benchmark results, no hyperparameters tuning was performed on the test set. Prompt design adhered to widely accepted practices, explicitly defining the model’s role, core capabilities, and task objectives. All experiments—including prompt design, model inference, and evaluation—were conducted in Chinese to reflect the authentic clinical linguistic environment. To evaluate the effectiveness and stability of the prompts, both automated and manual assessments were carried out on a small, independent development set, which included both objective and subjective tasks and excluded all test set instances. The finalized prompts were applied consistently across all models under evaluation. Advanced optimization techniques such as few-shot demonstrations, self-consistency, or iterative refinement were deliberately excluded to ensure that the evaluation results reflect the models’ inherent clinical reasoning capabilities.

Development set and test set

We constructed a dedicated development set to support prompt validation and future experiments involving few-shot learning or hyperparameter tuning. The complete GPBench dataset comprises 88 standardized clinical cases encompassing 8 top common chronic diseases and 10 most common symptoms, along with 4,011 multiple-choice questions accompanied by verified reference answers. To ensure rigorous evaluation, the data were split as follows:

- Development set:
 - *MCQ Dev Set*: 350 multiple-choice questions sampled using a stratified approach according to evaluation dimensions to ensure balanced coverage.
 - *Clinical Case Dev Set*: 18 cases randomly selected from the total 88, balanced by disease category and common symptoms, used exclusively for prompt validation and hyperparameter experiments.
 - *AI Patient Dev Set*: “AI patient” agents generated from the 18 clinical cases in the *Clinical Case Dev Set* following designed prompt instructions.
- Test set:
 - *MCQ Test Set*: The remaining 3,661 multiple-choice questions.
 - *Clinical Case Test Set*: The remaining 70 clinical cases, used exclusively for final evaluation.

- *AI Patient Test Set*: “AI patient” agents generated from the 70 clinical cases in the *Clinical Case Test Set* using the same prompt instructions.

This design ensures a strict separation between development and test sets, enabling prompt validation and controlled tuning without compromising benchmark integrity.

Supplementary Method 4. Supplementary materials for human evaluations

To provide a reference for interpreting LLM performance in GPBench, we conducted a human evaluation involving general practitioners (GPs). All testing procedures were fully aligned with those applied to the LLMs, ensuring direct comparability. A total of six GPs participated in the evaluation, divided into three groups of two participants each to represent varying levels of clinical experience: two with more than 10 years, two with 5–10 years, and two with less than 5 years of general practice experience, as detailed in Supplementary Table 10. All assessments were conducted independently in a quiet environment, without the aid of any external tools.

Supplementary Table 10. Basic information of the GPs who participated in the evaluation.

Group	The Number of Participants	Years of Working Experience in General Practice
A	2	> 10 years
B	2	5 ~ 10 years
C	2	< 5 years

The test materials consisted of three components, detailed as follows:

- *MCQ Test Set*: A total of 200 multiple-choice questions were selected from the complete MCQ test set using a stratified random sampling method under the guidance of clinical physicians, ensuring balanced coverage across all 14 competency dimensions. Participants completed the questions via a digital questionnaire, and accuracy was automatically calculated.
- *Clinical Case Test Set*: All 70 real-world outpatient cases were presented in a structured questionnaire format, requiring participants to provide free-text responses addressing both diagnosis and treatment plans.
- *AI Patient Test Set*: The same 70 cases were deployed as interactive simulated consultations on the Dify platform. Participants interacted with AI-generated “patients” using standardized prompts and scripted dialogues identical to those applied to the LLMs.

Before testing, participants received standardized instructions consistent with the prompts used for the LLMs. The responses of GPs to the *Clinical Case Test Set* and the *AI Patient Test Set* were manually evaluated by experienced general practitioners using the same scoring criteria and procedure. The detailed results of the human evaluations are presented in Supplementary Table 9.

Supplementary Method 5. Supplementary materials for the Clinical Case Test Set and the AI Patient Test Set

For the Diagnostic Plan, input the patient’s basic information, chief complaint, medical history, along with the results of physical and auxiliary examinations into the LLM. Based on the provided information, the LLM will assess whether a referral is necessary, determine if the case constitutes an emergency or critical condition, identify any existing complications, and produce a diagnostic result.

For the Treatment Plan, input the patient’s basic information, chief complaint, medical history, physical examination findings, auxiliary examination results, and the confirmed diagnostic result. The LLM will generate an optimal treatment plan based on this information and will additionally recommend further examinations, health education, and measures related to controlling treatment costs.

For patient agents in the *AI Patient test set*, the input to the LLM consists of the patient’s basic information, chief complaint, and medical history. The doctor agent, by contrast, receives no prior patient information and must acquire all relevant details through interaction with the patient agent. It should be noted that discrepancies may exist between the examination orders (both physical and auxiliary) issued by the doctor agent and the actual results. During the conversation, the true outcomes of the physical and auxiliary examinations are provided to the doctor agent. Based on the information gathered from the interaction and the examination results, the doctor agent formulates a diagnostic conclusion and proposes a treatment plan.

In the *Clinical Case Test Set* and *AI Patient Test Set*, the I3-1 indicator—*Formulation of Optimal Treatment Plan*—consists of two components: the optimal treatment plan and the clinical appropriateness evaluation of diagnostic examinations. Experts conducted fine-grained annotations for both components within the medical records. During evaluation, each component was independently scored by the three-member GP panel on a 100-point scale based on the expert-annotated standards. The final I3-1 score was then computed as the weighted average of the two component scores, with equal weighting applied.

Supplementary Data

- Supplementary Data 1. The competency indicators and definitions used in our proposed evaluation framework.
- Supplementary Data 2. An example of a CAD medical case.
- Supplementary Data 3. The performance (scores) of LLMs on the Clinical Case Test Set.

Supplementary References

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